you changed your API to use an external user store such as LDAP (discussed in the next section) or an OpenID Connect identity provider (chapter 7). In these cases, the groups that a user is a member of are likely to be returned as additional attributes during authentication (such as in the ID token JWT) rather than exist in the API's own database.

8.1.1 LDAP groups

In many large organizations, including most companies, users are managed centrally in an LDAP (Lightweight Directory Access Protocol) directory. LDAP is designed for storing user information and has built-in support for groups. You can learn more about LDAP at https://ldap.com/basic-ldap-concepts/. The LDAP standard defines the following two forms of groups:

1. *Static groups* are defined using the `groupOfNames` or `groupOfUniqueNames` object classes,[33] which explicitly lists the members of the group using the `member` or `uniqueMember` attributes. The difference between the two is that `groupOfUniqueNames` forbids the same member being listed twice.
2. *Dynamic groups* are defined using the `groupOfURLs` object class, where the membership of the group is given by a collection of LDAP URLs that define search queries against the directory. Any entry that matches one of the search URLs is a member of the group.

Some directory servers also support *virtual static groups*, which look like static groups but query a dynamic group to